Materials Science & Engineering A 556 (2012) 15–22

Contents lists available at SciVerse ScienceDirect

Materials Science & Engineering A

journal homepage: www.elsevier.com/locate/msea

Effects of W content on high temperature oxidation resistance and room temperature mechanical properties of hot-pressing Nb–XW alloys

Bairu Li [a,b], Shaosong Jiang [a,*], Kaifeng Zhang [a]

[a] National Key Laboratory for Precision Heat Processing of Materials, Harbin Institute of Technology, Harbin 150001, China
[b] College of Material Science and Engineering, Heilongjiang Institute of Science and Technology, Harbin 150027, China

ARTICLE INFO

Article history:
Received 20 January 2012
Received in revised form
6 May 2012
Accepted 16 June 2012
Available online 6 July 2012

Keywords:
Hot-pressing Nb–W alloys
W content
Oxidation resistance
Flexural strength
Fracture toughness

ABSTRACT

Oxidation behaviors at 1300 °C, room temperature flexural strength and fracture toughness of hot-pressing Nb–XW (X=10, 20, 30 wt%) alloys were investigated by isothermal oxidation, three-point bending (TPB) and un-precracked single edge notch bending (SENB) tests. It is shown that the oxidation resistance of Nb–XW alloys was improved with the increasing of W content due to WO_3 in oxidation scale reducing the oxygen-vacancy concentration and preventing Nb^{5+} from outward diffusing. However, 30% W was a limit concerning the effectiveness of W in improving oxidation resistance of Nb–W alloys. W addition to Nb gives rise to an increase in room temperature flexural strength due to solid solution strengthening, but W content must meet or even exceed 30% which can efficiently strengthen the alloy. Fracture toughness first increases obviously and then decreases slightly with W changing from 10% to 30%, which was attributed to ductile fracture propensity in 20 W alloy and remarkable solid solution strengthening in 30 W alloy, respectively. Considering the good balance of high temperature oxidation resistance, room temperature flexural strength and fracture toughness, W content of 30% was optimum for Nb–W alloys.

1. Introduction

Materials which could resist high temperatures and environmental corrosion are the relentless pursuit of the goal in the power generation and aerospace industries. Nb-based refractory alloys with a higher melting point and a relatively low density have been expected as candidate materials to use at ultra-high temperature over the maximum operating temperature of Ni-base superalloys. Researches on Nb-based refractory materials were constantly reported in many years [1–11]. Nb–W alloy can be widely used in aviation aerospace field because W with even higher melting point, as a main solution strengthening element, can effectively enhance high-temperature and room-temperature strength and appropriately improve high temperature oxidation resistance of Nb [12]. However, the fracture toughness of Nb–W alloy decreases with the addition of W due to the lower room-temperature plasticity and higher ductility brittleness transition temperature (DBTT) of W [13]. So, it is essential to consider the good balance between high temperature oxidation resistance and room temperature mechanical properties of Nb–W alloy. The oxidation behaviors [14–19] and mechanical properties [20–25] of Nb–W alloys with multicomponent systems have been studied by many scholars, respectively. However, the improving oxidation resistance of Nb–W alloy by way of alloying will inevitably reduce corresponding room temperature fracture toughness. In contrast, the applying high temperature oxidation resistant coating on Nb–W alloy will enhance the maximum operating temperature without lessening fracture toughness. In that situation, the fracture toughness and strength at room temperature become particularly important for Nb–W alloy as basis materials; meanwhile, the high temperature oxidation resistance should be also considered properly.

The aim of this research is to gain an understanding of the effects of W content on oxidation behaviors at high temperature, fracture toughness and flexural strength at room temperature of Nb–W alloy, and then to determine appropriate W content in Nb–W alloy which ensure the good balance of high temperature oxidation resistance and room temperature fracture toughness and flexural strength. Meanwhile, vacuum hot-pressing sintering (VHPS) was used to fabricate Nb–W alloy because the component segregation and the coarse microstructure of as-cast Nb-based alloy deteriorate the room temperature ductility and the oxidation resistance at high temperature [26–28].

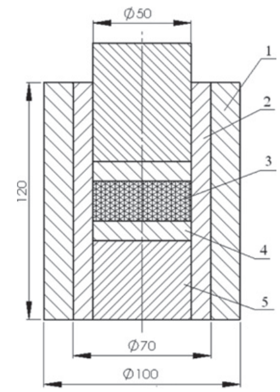
2. Experimental procedures

2.1. Alloys preparation and characterization

Mechanical alloying was carried out at room temperature in a planetary ball-milling machine (QM-BP) with stainless steel cell

* Corresponding author. Tel.: +860451 86413681; fax: +86451 86418763.
E-mail address: lbr-shmily@163.com (S. Jiang).

0921-5093/$ - see front matter © 2012 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.msea.2012.06.053



1- outer sleeve, 2- inner sleeve, 3- sintering materials, 4- gasket, 5- pressure head

Fig. 1. Dies drawing for hot-pressed sintering.

and bearing steel balls. The cell was loaded with mixture of Nb (99.9 wt%, −200 meshes) and W (99.8 wt%, −250 meshes) powders together with 500 g balls. The powders were blended in appropriate amounts to give the nominal composition of Nb–*X*W (*X*=10, 20, 30 wt%). Absolute ethyl alcohol was used as process control agents (PCA) during milling. During milling, the ball to powder weight ratio, solid to liquid ratio, milling time and rotation speed were maintained at 6:1, 1:0.2, 12 h and 250 rpm [29], respectively. Milled powders were screened and dried for 20 h at temperature of 90 °C in vacuum drying chest. Powder sintering was proceeded in the HIGH MULTI1000 type hot-pressing furnace. The powders were loaded into high-strength graphite mould shown in Fig. 1 and cold-pressed with pressures of 10 MPa at room temperature, and then uniaxially hot pressed under argon atmosphere. The mould was firstly raised to 1200 °C using a heating rate of 25 °C/min and kept 5 min, followed to 1800 °C using a heating rate of 20 °C/min and kept 90 min under 30 MPa, and then furnace cooled to ambient temperature. The sintering process was described in detail in Fig. 2.

Morphology and phase observation of the powders were carried out using S4700 scanning electron microscopy equipped with energy dispersive spectroscopy (SEM/EDS) and X-ray diffraction (XRD), respectively. The average sizes of powders were measured by the laser-diffraction diameter tester (IAC-05-001). Density of the hot-pressed samples was determined by the immersion method in distilled water, based on Archimedes principle. The densities were mathematical average, which were obtained from duplicate three specimens. Microstructure of the alloys was characterized using back scattered electron images (BSC) in SEM/EDS. Phases of the alloys were identified by XRD.

2.2. Oxidation experiments

The samples for oxidation of ϕ 6 mm × 9 mm were cut from the sintered samples with the dimensions of ϕ 50 mm × 9 mm using electro-discharge machining (EDM). After being ground to a final 600 SiC grit and ultrasonically cleaned in ethanol, then the samples were dried in air for using. Isothermal oxidation was carried out in air at 1300 °C in a box furnace using a heating rate of 10 °C/min by program control, the samples were held at the peak temperature for 0.5 h, 1 h, 2 h, 3 h and 4 h, respectively, and then furnace cooled to ambient temperature. Before and after oxidation experiments, the samples including the spallation scales were weighted using a balance with 0.01 mg sensitivity. XRD analysis was performed on the oxidized samples to identify constituent phases. Surface morphologies of the oxidation scales were characterized using secondary electron imaging (SEI) in SEM/EDS.

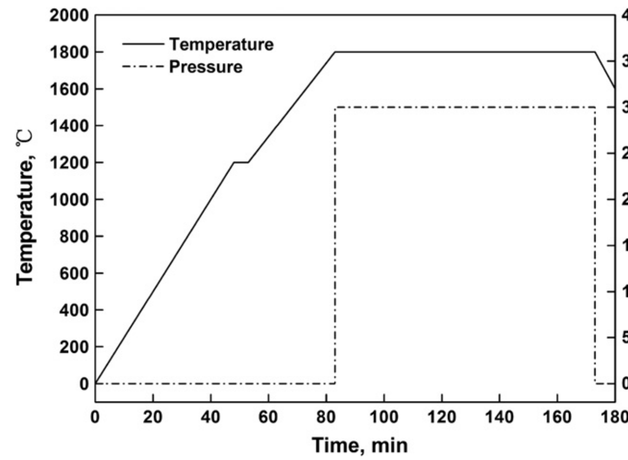


Fig. 2. Process route of hot-pressed sintering for Nb–*X*W alloys.

2.3. Mechanical tests

Flexural strength was determined by three-point bending (TPB) tests on samples with 30 mm span length (total length=36 mm), 4 mm width and 3 mm thickness in Instron-1186. Facture toughness was evaluated by un-precracked single edge notch bending (SENB) samples with 16 mm span length (total length=20 mm), 4 mm width and 2 mm thickness. A straight notch of about 2 mm depth was inserted into each specimen using EDM with 0.1 mm diameter wire. Observations of fracture surface were carried out using SEM.

3. Results and discussions

3.1. Microstructure of sintering materials

The typical morphology and XRD patterns of powders are shown in Fig. 3. The irregular blocks Nb (see Fig. 3a) and granular W (see Fig. 3b) powders were milled to distinctively exhibit the flake morphology (see Fig. 3c) and corresponding average sizes were about 30 μm. As shown in Fig. 3d, the XRD pattern of unmilled powders showed the presence of typical peaks of Nb and W. After milling, the intensities of the main diffraction peaks of Nb and W quickly reduced, but did not deviate from the standard peak. The result indicates that composite powders were unalloyed during milling, or could be partly alloyed due to the similar atomic radius of W (2.02 Å) and Nb (2.08 Å). After hot press sintering, the dense Nb–*X*W (*X*=10, 20, 30 wt%) sintering materials were obtained and corresponding densities were 98.25 ± 0.5%, 98.76 ± 0.3% and 99.01 ± 0.2%, respectively. Due to the limit of detection accuracy of balance, it is no much difference between three alloys. The microstructure and XRD patterns of sintering materials are shown in Fig. 4a and b, respectively. As shown in Fig. 4b, the peaks of W disappeared and those of Nb moved towards the high angle side with the increasing of W contents. The result shows that W as the replacement atoms was dissolved into Nb lattices to form α-Nb solid solution based on the Nb–W binary phase diagram. However, the positions of Nb peaks in 20 W alloy were almost same as those in 10 W, suggesting the unobvious lattice distortion appeared in 20 W alloy when W atoms were dissolved into Nb lattices due to the similar atomic radius of W (2.02 Å) and Nb (2.08 Å), which maybe have an effect on the room temperature strength of 20 W alloy. The BSE image shown in Fig. 4a indicates that there was no second phase or unexpected inclusion in Nb–*X*W alloys, indicating W was soluble completely in Nb lattices and formed α-Nb solid solution. The EDS

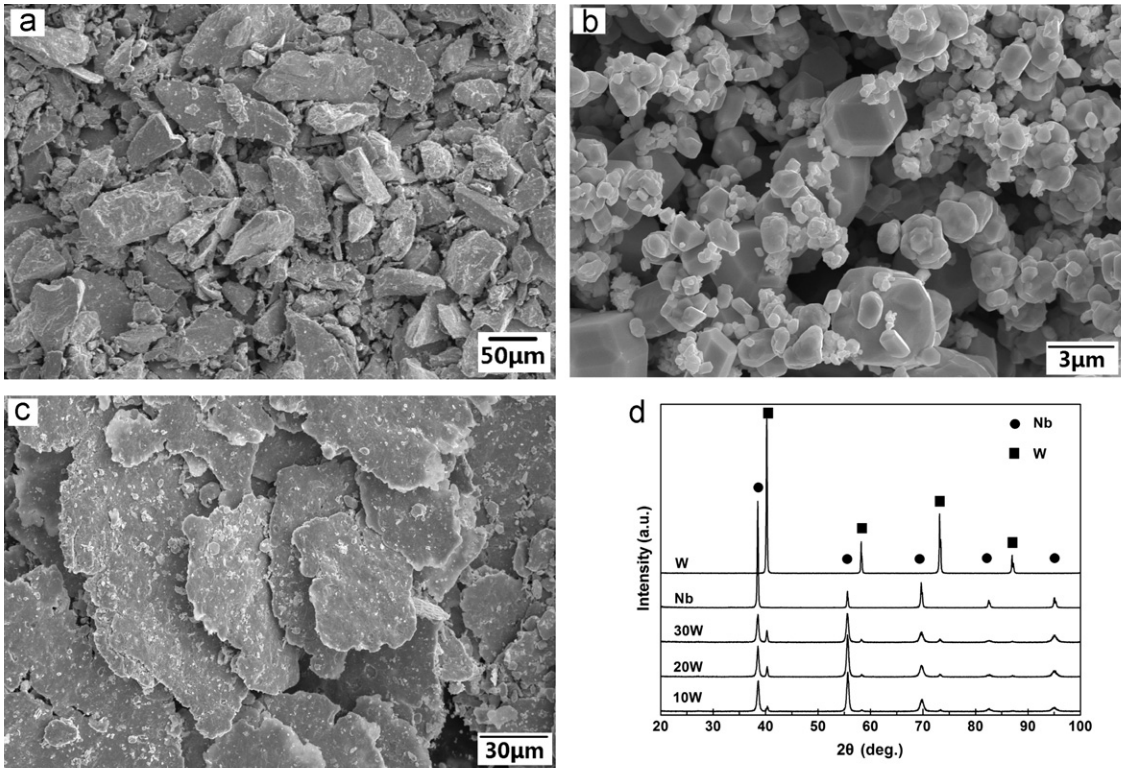


Fig. 3. Morphologies of (a) pure Nb, (b) pure W and (c) as-milled Nb–*X*W powders and (d) corresponding XRD patterns.

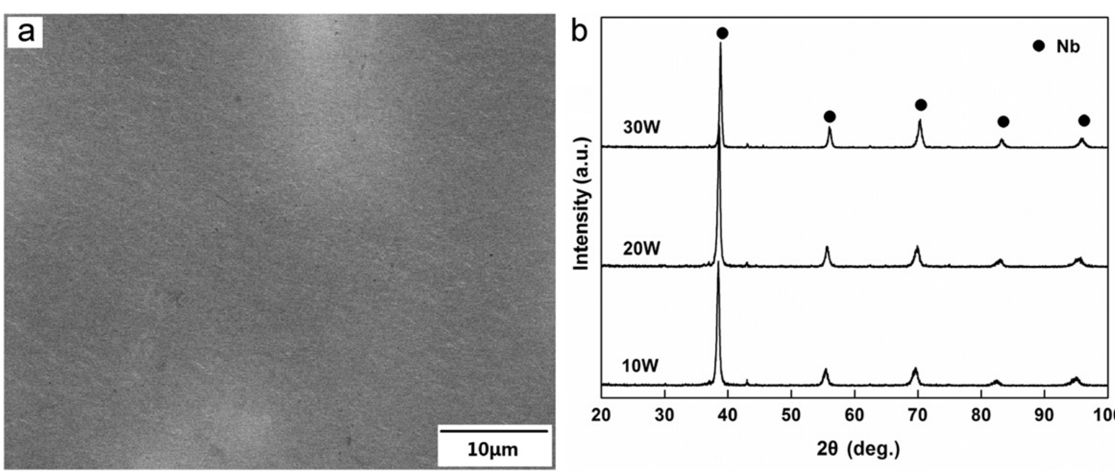


Fig. 4. BSE image (a) and XRD patterns (b) of hot-pressing Nb–*X*W alloys.

results of phase in Nb–*X*W alloys listed in Table 1 also confirmed this, which was accordance with XRD analysis.

3.2. *Oxidation resistance of sintering materials*

The oxidation rates of Nb–*X*W alloys were measured at 1300 °C in static air. Fig. 5 shows the mass change per unit area as a function of the oxidation time at 1300 °C for the 10 W, 20 W and 30 W alloys. The oxidation behavior of all the alloys followed a parabolic law at this temperature; the time at which the oxidation rate stabilizes depended on the alloy composition. The alloys with higher W contents exhibited better oxidation resistance at 1300 °C, the mass change per unit area of 10 W alloy was almost 2 times that of 30 W, delineating the relationship between the oxidation resistance and the W contents in the alloys. The large difference in mass change per unit area between alloys 10 W and 20 W and the small difference between alloys 20 W and 30 W suggest the existence of a limit concerning the effectiveness of the W in improving oxidation resistance of these alloys. The parabolic rate constants for oxidation tests are listed in Table 2. The parabolic oxidation rate constants (k_p) were calculated by least-square, linear-regression analysis of the mass gain per unit surface ($\triangle m/A$) and the exposure time (t):

$$\left(\frac{\Delta m}{A}\right)^2 = k_P t \tag{1}$$

According to Table 2, the parabolic rate constants for 20 W and 30 W alloys were almost exactly the same, which indicated that 30 W was a limit concerning the effectiveness of the W in improving oxidation resistance of Nb–*X*W alloys.

Table 1
Elemental compositions of phase in Nb–XW alloys determined by EDS analysis.

Alloys	Compositions (wt%)	
	Nb	W
Nb–10 W	90.31	9.69
Nb–20 W	80.68	19.32
Nb–30 W	71.93	28.07

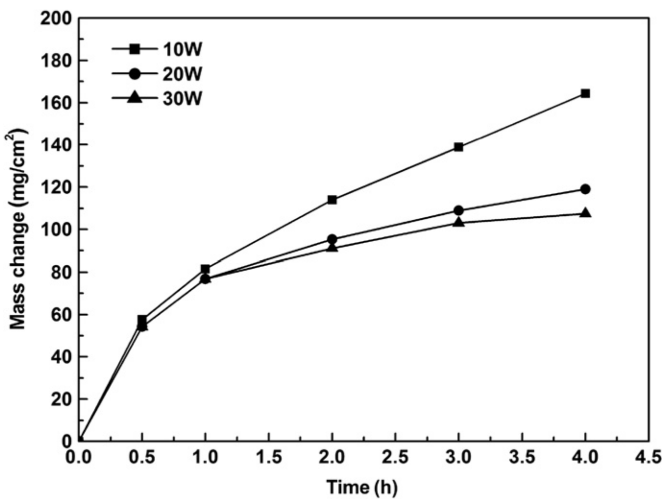


Fig. 5. Cyclic oxidation curves for samples 10 W, 20 W and 30 W at 1300 °C.

Table 2
Values of parabolic rate constants (k_p) and correlation coefficients corresponding to cyclic exposure at 1300 °C.

Alloys	Time span (h)	k_p ($mg^2\,cm^{-4}\,h^{-1}$)
10 W	0–4	6640.62
20 W	0–4	5893.63
30 W	0–4	5884.42

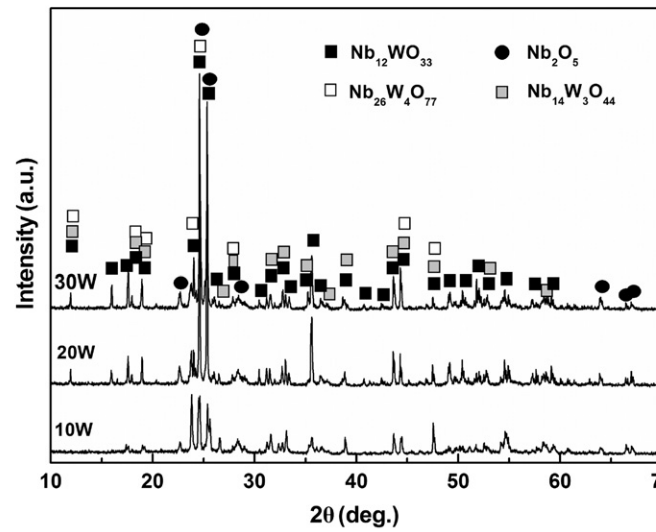


Fig. 6. XRD patterns of the oxidation scales formed on Nb–XW alloys after 4 h oxidation at 1300 °C.

Fig. 6 shows the XRD patterns of oxide scales formed on Nb–XW alloys. Monoclinic $Nb_{12}WO_{33}$, monoclinic $Nb_{26}W_4O_{77}$, $Nb_{14}W_3O_{44}$ monoclinic and Nb_2O_5 were identified on the oxide scales, $Nb_{12}WO_{33}$, $Nb_{26}W_4O_{77}$ and $Nb_{14}W_3O_{44}$ of which were different (Nb_2O_5+WO_3) solid solution due to the dissolution of WO_3 in Nb_2O_5 [30]. WO_3 was not detected by XRD which was attributed to a large solid solubility of WO_3 in Nb_2O_5. As shown in Fig. 6, the increase of (Nb_2O_5+WO_3) solid solution with the increasing of W content, which maybe improve the oxidation resistance of Nb–W alloy. Fig. 7 shows the surface morphologies and corresponding EDS patterns of oxide scales formed on Nb-XW alloys after oxidation at 1300 °C for 1 h and 4 h. As shown in Fig. 7a–c, the oxides of these three alloys all presented fine granular mixed with columnar appearances after oxidation at 1300 °C for 1 h. Moreover, the oxide scales were denser and mostly consisted of (Nb_2O_5+WO_3) solid solution based on EDS results and XRD analysis, indicating the similar oxidation resistance after oxidation for 1 h. The result was accordance with that described by the cyclic oxidation curves of samples, suggesting the effect of W contents on oxidation resistance was insignificant when the oxidation time was shorter. However, the positive influence of W contents on the improving of oxidation resistance was embodied with the extending of oxidation time. After oxidation for 4 h, the oxide scales still were composed of (Nb_2O_5+WO_3) solid solution based on EDS results. Although the oxides in 10 W alloy still exhibited fine granular, the oxides scale was very loose suggesting the poor oxidation resistance, as shown in Fig. 7d. However, 20 W and 30 W alloys exhibited the better oxidation resistance due to the denser oxides scale though the fine granular oxides grew into coarse columnar, as shown in Fig. 7e and f. The reason may be due to the increasing of (Nb_2O_5+WO_3) solid solution in the oxides scale according to the EDS results and XRD analysis, which was described in detail as addressed bellow [31]. At the onset of oxidation, Nb started to oxidize preferentially to Nb_2O_5 by inward diffusion of oxygen, which caused the W concentration at the metal/oxide interface increase. W in the alloys reacted with inward diffusing oxygen to form WO_3 oxides when reaching a critical concentration. Moreover, WO_3 increased with the increasing of W contents and were gradually dissolved into Nb_2O_5 leading to the formation of (Nb_2O_5+WO_3) solid solution. The diffusion rate of Nb^{5+} through WO_3 or (Nb_2O_5+WO_3) solid solution was considerably less than through Nb, which could have a blocking effect on the outward diffusion of Nb^{5+}, meanwhile, the oxygen-vacancy concentration decreased due to the increase of WO_3[16]. Afterward, the Nb_2O_5 growth was completely prevented, leading to the improvement of the oxidation resistance.

3.3. Mechanical properties of sintering materials

Fig. 8a and b shows the flexural strength and fracture toughness at room temperature as a function of W contents in Nb–XW alloys, respectively. The flexural strength increased with the increasing of W content, as seen in Fig. 8a, which may be closely associated with solid solution strengthening and diffusivity of W. The effect of grain sizes on the flexural strength was negligible, because the grain sizes of these three alloys were all about 30 μm measured by the fracture surfaces of TPB samples with the transgranular cleavage shown in Fig. 9, which were not sensitive to W content. Moreover, the lattice parameter of Nb solid solution decreased with the increasing W content, as shown in Fig. 4b. Therefore, solid solution strengthening due to elastic interaction arising from the atomic size difference may be one of the important strengthening mechanisms at room temperature in the single-phase bcc alloys [21]. However, the trend of flexural strength was different with the increasing of *W* content: the

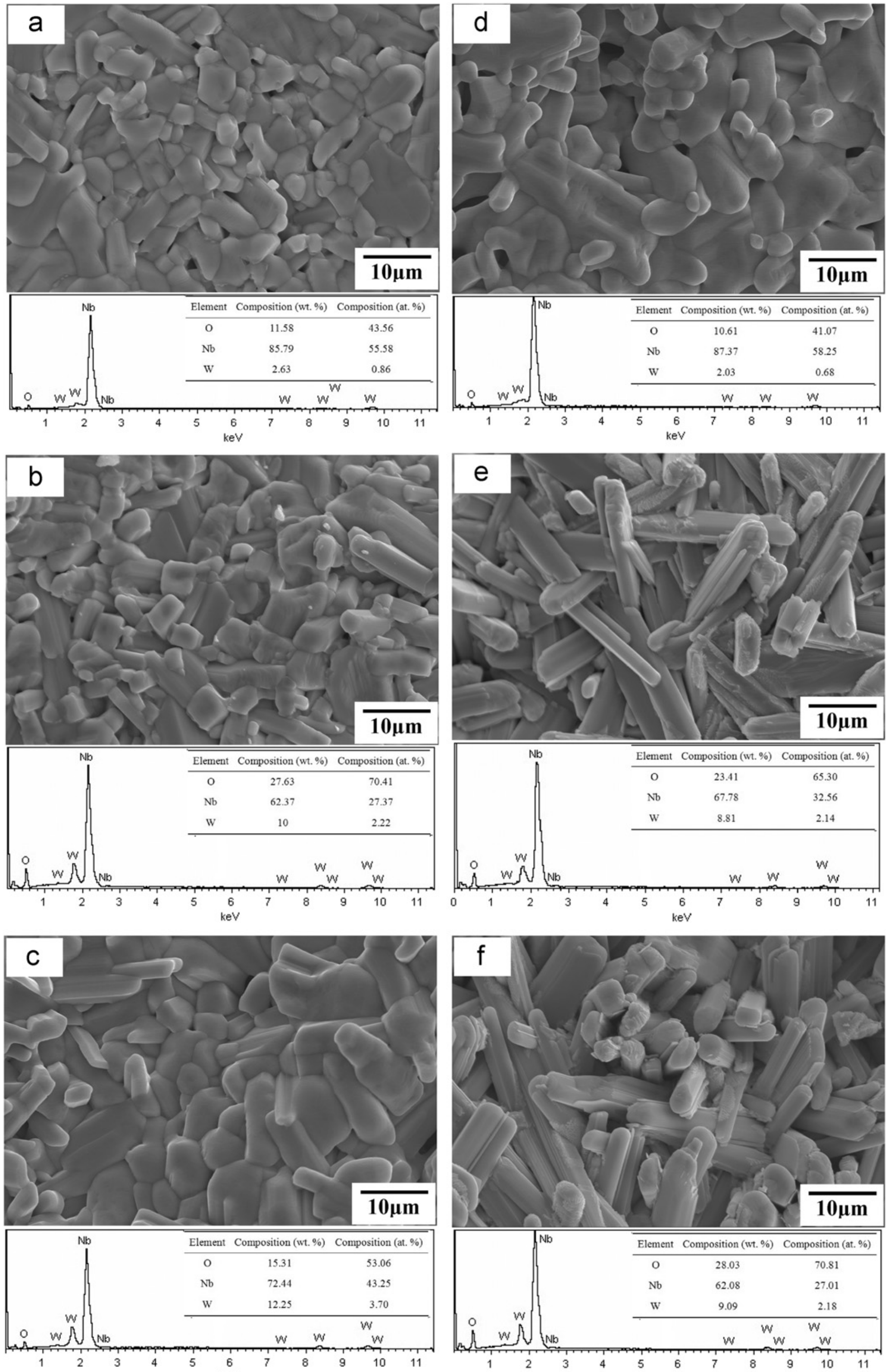


Fig. 7. Oxidation scales and corresponding EDS of (a) 10 W, (b) 20 W and (c) 30 W alloys oxidized for 1 h and (d) 10 W, (e) 20 W and (f) 30 W alloys oxidized for 4 h at 1300 °C.

flexural strength increased slightly with the W content changing from 10% to 20% and increased rapidly from 20% to 30%. The reason may be due to as addressed below. The lattice parameters of Nb solid solution for 10 W and 20 W alloys were almost exactly the same, as shown in Fig. 4b, suggesting the lattice distortion caused by W dissolved into Nb lattices had no obvious changes due to the similar atomic radius of W (2.02 Å) and Nb (2.08 Å). Accordingly, 20 W alloy could not be strengthened efficiently.

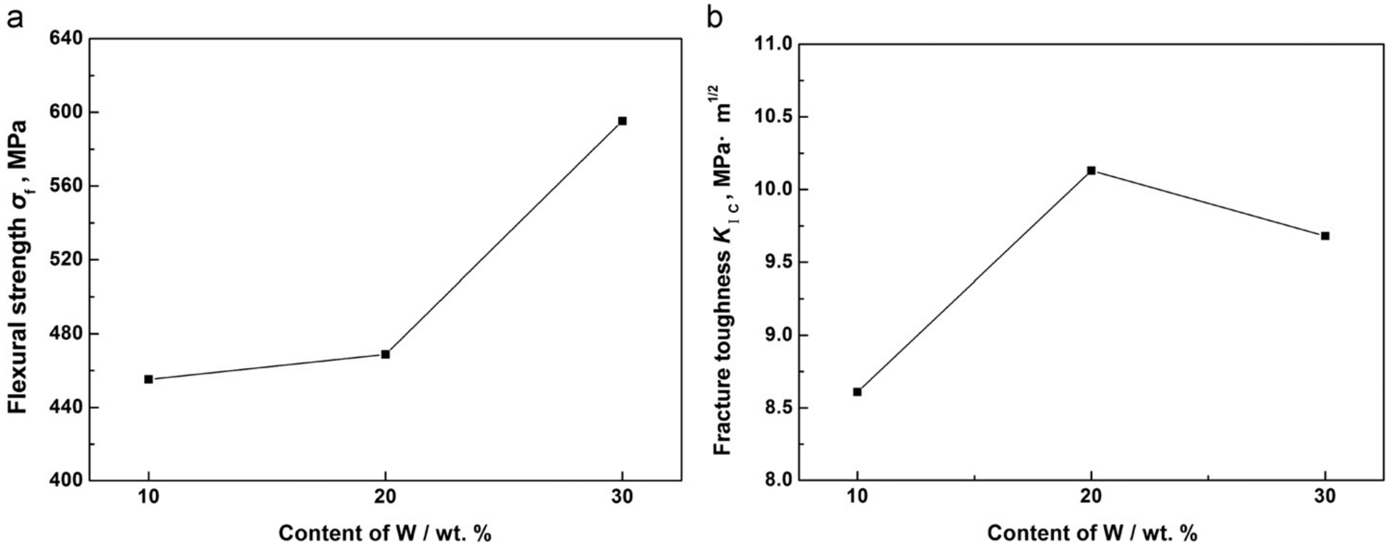


Fig. 8. Relationship of (a) room temperature flexural strength and (b) fracture toughness plotted as a function of W content in Nb–XW alloys.

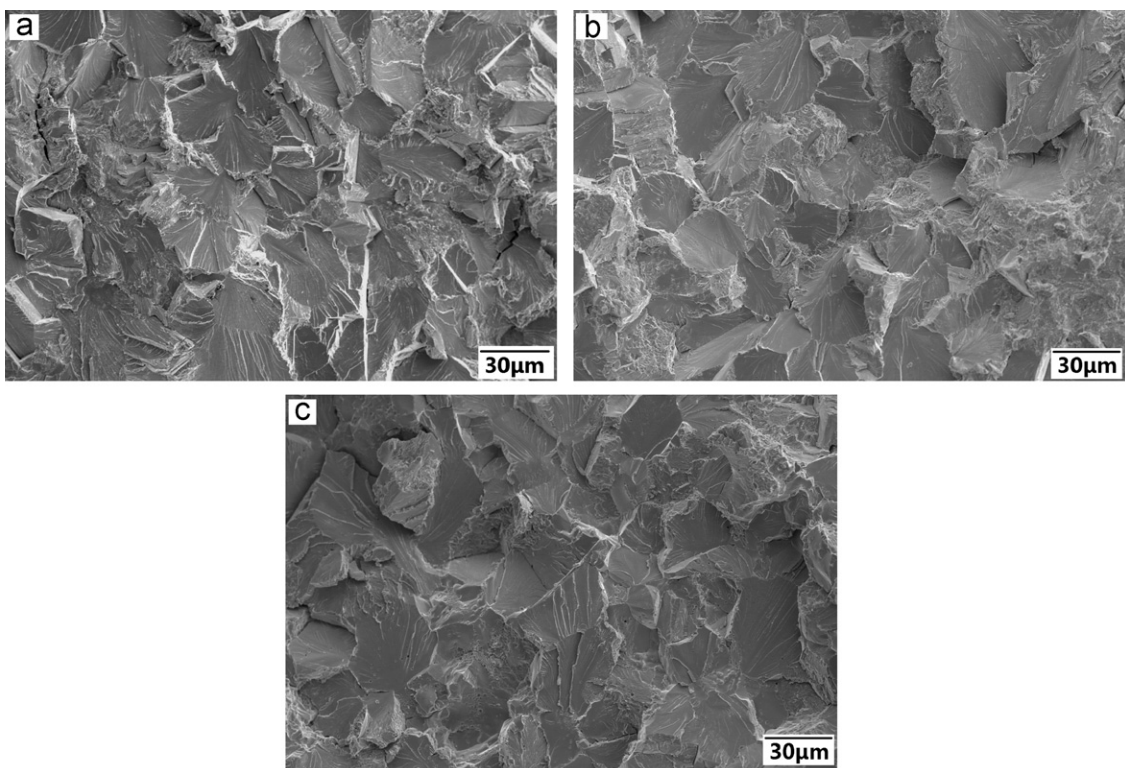


Fig. 9. Fracture surfaces of TPB samples taken from (a) 10 W, (b) 20 W and (c) 30 W alloys.

However, the lattice parameter of Nb solid solution in 30 W alloy was much smaller than that of 20 W alloy, indicating the serious lattice distortion which greatly strengthen 30 W alloy. As a result, a conclusion can be drew that the W content must reach or exceed 30% which can strengthen the alloy efficiently as a main solution strengthening element of Nb–W alloy.

As shown in Fig. 8b, the fracture toughness increased rapidly with the W content changing from 10% to 20% and decreased a little from 20% to 30%, which was attributed to the ductile fracture propensity in 20 W alloy (see Fig. 10b) and the remarkable solid solution strengthening in 30 W alloy, respectively. Fig. 10 shows the fracture surfaces of SENB samples taken from (a) 10 W, (b) 20 W and (c) 30 W alloys. In the 10 W alloy, fracture occurred mostly in a transgranular cleavage manner, while in the 20 W and 30 W alloys, honeycomb structure and transgranular cleavage were observed on the whole fracture surface. Therefore, the difference in fracture toughness among 10 W, 20 W and 30 W alloy would be explained in terms of the fracture mode. Although W addition of 20% could not efficiently strengthened 20 W alloy, it increased plastic zone at a crack tip, thereby enhancing ductile fracture and leading to the increase in fracture toughness. However, 30% W greatly strengthen 30 W alloy, which caused cleavage fracture to be in dominant and reduced fracture toughness. The reason and mechanism of change of fracture mode, especially the

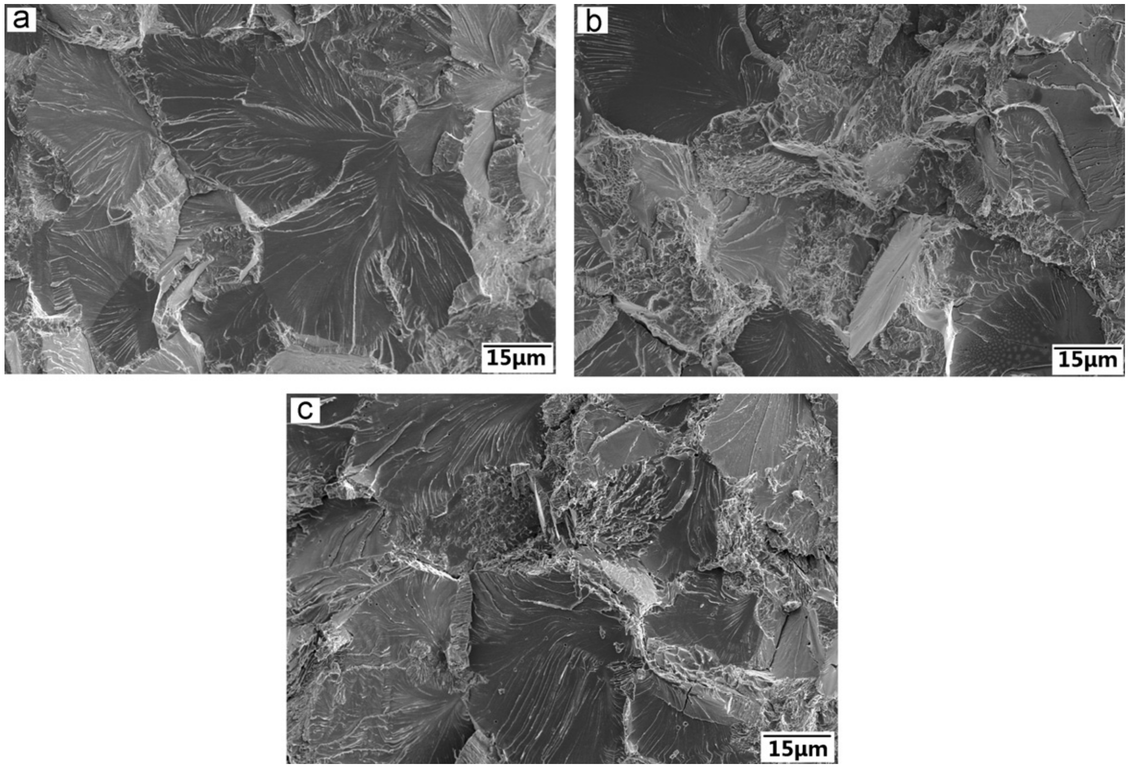


Fig. 10. Fracture surfaces of SENB samples taken from (a) 10 W, (b) 20 W and (c) 30 W alloys.

increase of plastic zone with the increase of W content were not clear, which need to be further investigated. However, based on above result, W content must be controlled within 20% to 30%, which can efficiently improve the fracture toughness of Nb–W alloy.

According to above analyses, these three facts consisting of high temperature oxidation resistance, room temperature flexural strength and fracture toughness taken together, W content of 30% should be optimum for Nb–W alloys.

4. Conclusions

Nb–XW (X=10, 20, 30 wt%) refractory alloys were fabricated by vacuum hot pressed sintering and corresponding high temperature oxidation resistance at 1300 °C, room temperature flexural strength and fracture toughness were investigated as a function of W content. The results obtained are summarized as follows.

(1) The Nb–W alloy exhibited better oxidation resistance at 1300 °C with the increasing of W contents. And 30 wt% was a limit concerning the effectiveness of W in improving oxidation resistance of Nb–W alloys.
(2) The flexural strength increased with the increasing of W content. However, W content must reach or even exceed 30 wt% which can strengthen Nb–W alloy efficiently.
(3) The fracture toughness increased rapidly with W content changing from 10% to 20% and decreased a little from 20% to 30%, indicating W content must be controlled within 20% to 30% which can efficiently improve the fracture toughness of Nb–W alloy.
(4) Considering the good balance of high temperature oxidation resistance and room temperature flexural strength and fracture toughness, W content of 30 wt% was optimum for Nb–W alloys.

Acknowledgements

Support from the National Natural Science Foundation of China (Grant No. 50775052) is acknowledged.

References

[1] D.M. Shah, D.L. Anton, D.P. Pope, S. Chin, Mater. Sci. Eng. A 192/193 (1995) 658–672.
[2] P.R. Subramanian, J.A. Shields, JOM 48 (1) (1996) 32.
[3] P.R. Subramanian, M.G. Mendiratta, D.M. Dimiduk, JOM 46 (1) (1996) 33–38.
[4] M.R. Jackson, JOM 48 (1) (1996) 39–44.
[5] D.Z. Yao, R. Cai, C.G. Zhou, J.B. Sha, H.R. Jiang, Corros. Sci. 51 (2009) 364–370.
[6] J. Wang, X.P. Guo, J.M. Guo., Chin. J. Aeronaut. 22 (2009) 544–550.
[7] R.G. Ding, I.P. Jones, H.S. Jiao, Mater. Sci. Eng., A 485 (2008) 92–98.
[8] M. Yoshida, T. Takasugi, Mater. Sci. Eng., A 224 (1997) 69–76.
[9] J. Genga, P. Tsakiropoulos, G.S. Shao, Mater. Sci. Eng., A 441 (2006) 26–38.
[10] B.I. Portillo, S.K. Varma, J. Alloys Compd. 497 (2010) 68–73.
[11] V. Behrani, A.J. Thom, M.J. Kramer, M. Akinc, Intermetallics 14 (2006) 24–32.
[12] M. Nedyakha, V.G. Chernyi, Fiziko Khimicheskaya Mekhanika, Materialov 2 (6) (1966) 646–648.
[13] B.S. Lement, I. Perlmutter, J. Less-Common Met. 2 (1960) 253–271.
[14] C.T. Sims, W.D. Klopp, RI Jaffee, Transations 15 (1959) 257.
[15] GL Miller, FG Cox, J. Less-Common Met. 2 (1960) 207–222.
[16] V.K. Sikkai, C.J. Rosa, Corros. Sci. 20 (1980) 1221–1239.
[17] Maria del Pilar Moricca, S.K. Varma, Corros. Sci. 52 (2010) 2964–2972.
[18] Maria del Pilar Moricca, S.K. Varma, J. Alloys Compd. 489 (2010) 195–201.
[19] Benedict Portillo Julieta Ventura, S.K. Varma, J. Alloys Compd. 476 (2009) 257–262.
[20] W.Y. Kim, H. Tanaka, A. Kasama, S. Hanada, J. Alloys Compd. 333 (2002) 170–178.
[21] W.Y. Kim, H. Tanaka, M.S. Kim, S. Hanada, Mater. Sci. Eng., A 346 (2003) 65–74.
[22] J.B. Sha, H. Hirai, T. Tabaru, A. Kitahara, H. Uenoa, S. Hanada., Mater. Sci. Eng., A 364 (2004) 151–158.
[23] P. Zheng, J. Sha, D. Liu, S. Gong, H. Xu, Mater. Sci. Eng., A 483–484 (2008) 656–659.

[24] D. Wang, X. Zheng, Z. Li, T. Zhang, R. Bai, M. Xia, J. Fu, Rare Met. Mater. Eng. 37 (2008) 618–621.
[25] X. Zheng, S.K. Chen, Z.K. Li, T.J. Zhang, J. Fu, D.H. Wang, Rare Met. 3 (2009) 1–5.
[26] M.R. Jackson, B.P. Bewlay, R.G. Rowe, JOM 1 (1996) 39–44.
[27] L. Yin, D.Q. Yi, L.R. Xiao, Mater. Prot. 36 (2003) 4–8.
[28] J.D. Ringty, P.M. Singh, J.J. Lewandowski, JOM 48 (1992) 36–41.
[29] B.R. Li, K.F. Zhang. Manuscript.
[30] Goldschmidt, Metallurgic 62 (1960) 211–241.
[31] B.R. Li, K.F. Zhang, Vacuum 86 (2012) 1341–1346.